INTERNATIONAL MONETARY FUND

TUVALU

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Approved By
Brian Aitken and Vikram Haksar (IMF), and Satu Kahkonen (IDA)

Prepared jointly by the staffs of the International Monetary Fund and the International Development Association

This year's Debt Sustainability Analysis (DSA) concludes that Tuvalu is at a high risk of debt distress, the same risk rating assigned in the previous DSA conducted in 2012. While a large fiscal surplus was recorded over the last two years, allowing the country to build up needed fiscal buffers, the increase in expenditures envisaged in the 2014 budget—financed by a temporary but large increase in foreign grants— implies a significant increase in the risk to fiscal sustainability. Under the baseline scenario, where the increase in the spending envelope is not scaled back in line with the unwinding of foreign grants in a timely manner, the budget would slip into a deficit from 2015 onwards. Although the PV of debt to GDP ratio would initially fall below the indicative policy threshold as fiscal buffers are depleted, over the medium term debt would build up again as fiscal buffers are exhausted, and reach above the indicative threshold in outer years of the projection. This trajectory underscores why a modest fiscal surplus is needed to lower the risk of debt distress to moderate levels. Also, placing growth on a higher trajectory by addressing structural weaknesses should strengthen the country's ability to service debt.

BACKGROUND

1. Being one of the smallest and most isolated countries in the world, Tuvalu faces tremendous challenges. With a population of some 11,000 people living on 26 square kilometers, Tuvalu is more than 3,000 kilometers away from its nearest major external market. Despite substantial, albeit volatile, foreign aid and a boom in the fisheries sector, growth prospects are hampered by weak competitiveness and capacity, with remoteness causing high transportation cost and isolation, and the lack of scale increasing transaction cost.

2. The authorities are cognizant of the challenges and have developed a broad reform agenda. A development strategy and reform package, which were formulated in consultation with development partners, have been fleshed out in an effort to enhance governance and social development, facilitate private sector growth, and safeguard macroeconomic stability.

3. The budget achieved a substantial surplus for the second consecutive year in 2013, but a dramatic increase of budget spending in 2014 has called into question its fiscal sustainability. The fiscal surplus reached 26.3 percent of GDP, reflecting record high fishing license fees and higher than expected tax revenue and foreign aid. Nevertheless, the 2014 budget targets a more than 20 percent expansion of spending, which would be financed by a large temporary increase of foreign grants and transfers. The substantial fiscal expansion would result eventually in a depletion of fiscal buffers and accumulation of foreign debt.

4. Meanwhile, the government's external assets have risen markedly. The Consolidated Investment Fund (CIF), which serves as a fiscal buffer and is fully controlled by the government of Tuvalu, has been built up to more than $A 15 million (38 percent of GDP). The Tuvalu Trust Fund (TTF), which was capitalized by development partners in 1987, has grown from $A130 million in 2012 to more than $A 140 million in 2013 (356 percent of GDP). The TTF is not fully sovereign and is owned instead by Tuvalu, Australia, and New Zealand, amongst others countries. It cannot be drawn down freely: when its market value exceeds a "maintained value" indexed to Australian CPI, the TTF Board—representing both development partners and Tuvalu government—can decide to transfer the funds in excess of the target income to the CIF. This DSA analyzes gross public debt because of the limited conditional access to the TTF.

5. Since the previous DSA conducted in 2012, public and publicly guaranteed (PPG) debt has continued to decline, but is expected to rise as a result of public enterprise borrowing in the near term. As of end-2013, total PPG debt stood at US$ 14.6 million, equivalent to 41 percent of GDP. External debt was US$ 12.5 million (35 percent of GDP), and domestic debt—owed to domestic banks—stood at US$ 2.1 million (6 percent of GDP). Bilateral development partners provide only grant assistance, while multilateral institutions, most notably the Asian Development Bank, provided concessional loans to the government. Meanwhile, loans contracted at nonconcessional terms by two fishing joint ventures established by the National Fishing Corporation

of Tuvalu (NAFICOT) and Asian companies account for a significant share of public debt. These loans are largely guaranteed by the government, and constitute contingent government liabilities. [1] Recently, one joint venture has been working on a new loan agreement to finance the expansion of its fishing capacity, which implies a large addition of public debt in the near term—most likely in 2014 or 2015.

Tuvalu: Public Debt, End-2013

	Creditor	Outstanding amount (US$ million)	Outstanding amount (Percent of GDP)	Concessional
External Debt		**12.5**	**35.3**	
Central Government Debt	ADB	6.2	17.4	Yes
Development Bank Debt	EIB	1.4	3.8	No
Fishing Joint Venture Debt	Commercial bank	5.0	14.1	No
Domestic debt		**2.1**	**5.8**	No

6. This DSA is built on a baseline macroeconomic framework that assumes an accommodative and procyclical fiscal policy. Without a timely action to cut back the spending envelope from the 2014 budget level, the fiscal balance would shift into a deficit from 2015 onwards, which would be financed initially by drawing down the CIF, and later, when the CIF is fully depleted in 2021, by concessional borrowing. This would cause debt-to-GDP ratio to decline over the medium term as the current debt is amortized, and build up again as new borrowing occurs. Although weaknesses in the banks and public enterprises could pose large liabilities to the government, these costs are not assumed in the baseline because of the lack of reliable data—the cost of addressing banking fragilities needs to be assessed on the basis of a prudent regulation and resolution framework, while the fiscal cost associated with public enterprise weaknesses should be estimated in the context of a comprehensive public enterprise reform package.

7. Compared to the previous DSA, the main features of the current macroeconomic framework can be summarized as follows:

- **Growth**. Real GDP is projected to grow by an average of 2.2 percent per annum over the medium term and 2 percent over the long run, compared to a constant of 1.2 percent in the previous DSA.

- **Inflation**. Inflation is projected to reach about 3 percent in the near term and gradually moderate to below 2 percent in the long run, which is similar to the previous DSA.

[1] An effectively defunct and wholly government-owned public enterprise called the National Fishing Corporation of Tuvalu (NAFICOT) is party to the two fishing joint ventures (JV). NAFICOT owns 50 percent of the JV that was established together with a company of the Taiwan Province of China, and 40 percent of the Korean one.

- **Balance of payments.** Exports of goods and services are projected to remain stable at 60 percent of GDP over the medium and long run compared to just above 10 percent in the previous DSA, with the difference reflecting revision to the data to account for large exports by fishing joint ventures following a recent technical assistance on external account statistics. Imports of goods and services would peak at 168 percent of GDP in the medium term and moderate to 115 percent of GDP over the long run; in the previous DSA, imports would peak at 151 percent of GDP over the medium term, and moderate to 112 percent of GDP over the long run. The current account deficit would widen to 37 percent of GDP over the medium term, and then narrow to 4.4 percent of GDP by the end of the projection period, a level close to the last DSA.

- **Fiscal.** Revenue is slightly higher than in the last DSA over the medium term, reflecting buoyant fishing license fees. However, it is expected to fall to 45 percent of GDP towards the end of the projection period, compared to 56 percent in the last DSA. Grants as a share of GDP will also fall over time, reaching 13 percent of GDP by 2034 compared to 17 percent of GDP in the previous DSA. Expenditures are projected to decline from 100 percent of GDP in 2014 to 62 percent in 2034, while the last DSA projected expenditure to be broadly stable at above 80 percent of GDP.

- **Discount rate.** The discount rate has been increased from 3 percent in the last DSA to 5 percent, reflecting the most recent DSA guidelines.

Tuvalu: Key Macroeconomic Assumptions
(In percent)

	2012 DSA	2014 DSA
	2012-2017 AVG	2014-2019 AVG
Real GDP growth	1.2	2.2
Inflation (GDP deflator)	1.9	2.1
Current account balance/ GDP	0.5	-11.9
Revenue excluding grants/ GDP	58.5	60.0
Grants/GDP	13.7	24.4
Expenditure/GDP	81.0	87.0
Fiscal Balance/GDP	-8.8	-2.2

8. Risks to the baseline scenario are on the downside. Vulnerabilities in state-owned banks and public enterprises imply large government liabilities. Uncertainties within the fisheries sector could pose additional risks to revenue and fiscal soundness. Moreover, a slowdown in advanced and emerging economies, including China, could undermine activity and reduce job opportunities in the Pacific region, which would weaken the demand for Tuvaluan seafarers and seasonal workers, adversely affecting Tuvalu's growth prospects.

DEBT SUSTAINABILITY

A. External Debt

9. Under the baseline scenario, external debt breaches the indicative threshold of debt-to-GDP ratio for prolonged periods. The threshold for PV of Debt to GDP is determined by the CPIA index, a measure of the country's institutional capacity. While the trajectory shows a decline in debt levels from 52 percent of GDP to just above 20 percent over the medium term, reflecting amortization of existing loans and no new borrowing in the medium term, increased borrowing only

in the outer years, leading to the full depletion of CIF buffers by 2021. This implies that the debt trajectory begins to rise again and breaches the threshold by 2029. (Figure 1) The PV of debt-to-GDP ratio starts at a much higher level than the last DSA, but converges to the level as the last DSA toward the end of the projection period. The negative contribution of residuals (table 1 and 2) to the debt trajectory throughout the medium-term horizon is largely driven by withdrawing from external assets under the CIF.

10. Stress tests show the debt ratios are highly sensitive to exports and exchange rate shocks. A one-time 30 percent nominal depreciation in 2015 would cause the debt-to-GDP ratio to stay above the indicative threshold for most of the projection period. A standard shock to exports also results in a breach of the debt-to-export threshold over the next few years. Also, the threshold for the debt service-to-exports would be breached for four years over the medium term.

B. Public Debt

11. The PV of public debt to GDP stays above the benchmark over the medium term. The PV of debt to GDP stands at 58 percent of GDP in 2014, and while declining gradually below the benchmark of 38 percent over the medium term, would rise again in outer years as new borrowing occurs after the CIF is depleted, reaching 40 percent of GDP by 2034.[2] (Figure 2). Although the breaching of the benchmark is small, it needs to be interpreted with caution. Since domestic banks are struggling with large amounts of nonperforming loans and can hardly extend additional financing on a sustained basis, external borrowing will constitute the main source of public financing, and debt sustainability analysis based on external debt ratios appears more informative at this stage. The PV of debt-to-revenue ratio would rise from 50 percent in 2014 to 69 percent in 2034, while the PV of debt service-to-revenue ratio will fall from 8 percent to 3 percent.

12. Paths of debt indicators are sensitive to alternative assumptions and bound tests. A one-time 30 percent depreciation would push the PV of debt-to-GDP ratio to peak at 73 percent in 2015. A standard shock to growth would cause the PV of debt-to-revenue ratio to rise from 50 percent in 2014 to 194 percent in 2034.

C. External and Public DSA under Alternative Scenario

13. Under an alternative scenario where fiscal policy is anchored by an appropriate fiscal balance target, the risk of debt distress is reduced considerably. Given the volatility of fishing license fees, targeting a structural balance would help reduce procyclicality—the structural balance is estimated on the basis of a moving average of fishing license fees over a period of seven years; fishing license fees in excess of the structural level, i.e. their moving average, would be saved in the CIF, while it can be drawn down in case fishing license fees fall short of the structural level. In the case of Tuvalu, targeting a moderate structural surplus implies saving fishing revenues in the current period given the dramatic increase in fishing license fees in the past few years. Under the alternative scenario, the fiscal policy targets a structural surplus of 0.5-1 percent of GDP over the medium to

[2] The public debt benchmark is derived on empirical basis, and varies among countries with their respective CPIA score.

long run. As a consequence, CIF buffers would be maintained at a comfortable level, and debt indicators will improve significantly—the PV of external debt-to-GDP would fall from 53 percent in 2014 to around 10 percent by 2034.

D. Authorities' View

14. The authorities broadly concurred with the overall assessment of the Debt-Sustainability Analysis, but made a couple of observations. First, the authorities, while recognizing the volatility of fishing revenues, are more optimistic about the near-term outcome, and expect revenues to be higher than staff projection in the next few years. Second, they indicated that the fishing joint venture has a favorable outlook, and the probability for the government to assume the company's debt obligation is small.

E. Conclusions

15. The results of Tuvalu's DSA point to a high risk of debt distress, which is the same conclusion that was drawn in the 2012 DSA. The PV of debt-to-GDP ratio is above the indicative threshold over the long run. Furthermore, a significant portion of public debt was contracted on nonconcessional terms, placing further strains on the country's weak debt servicing capacity.

16. Stress tests show that Tuvalu's debt dynamics are susceptible to shocks. Standard shocks in the DSA, particularly the one-time 30 percent nominal depreciation, as well as slower real GDP growth rates and lower export growth, would cause external debt indicators to breach their respective thresholds over the long run. Public debt indicators also worsen dramatically under bound tests.

17. The risk of debt distress is reduced to a moderate level under the alternative scenario where a modest fiscal structural surplus is implemented. Assuming that the government maintains a structural surplus of 0.5-1 percent of GDP over the medium and long term, all external debt indicators would decline below their indicative thresholds.

Box 1. Macroeconomic Assumptions Under Baseline Scenario

Growth and Inflation

- Growth is expected to be boosted to about 2.5 percent in the next few years thanks to the implementation of a few large infrastructure projects financed by grants from development partners.[1/] Over the long term, the growth will moderate to about 2 percent as these projects are completed. Meanwhile, the share of the public sector in the economy would gradually decline, while the private sector would play a more important role over time on the back of great potential in the fisheries sector, which is consistent with the Te Kakeega II development strategy.
- Inflation is projected to rise to about 3 percent in 2014 as a result of increased government spending and civil servant wages as well as a weakening Australian dollar. Inflation would moderate gradually to just below 2 percent in the outer years.

Balance of Payments

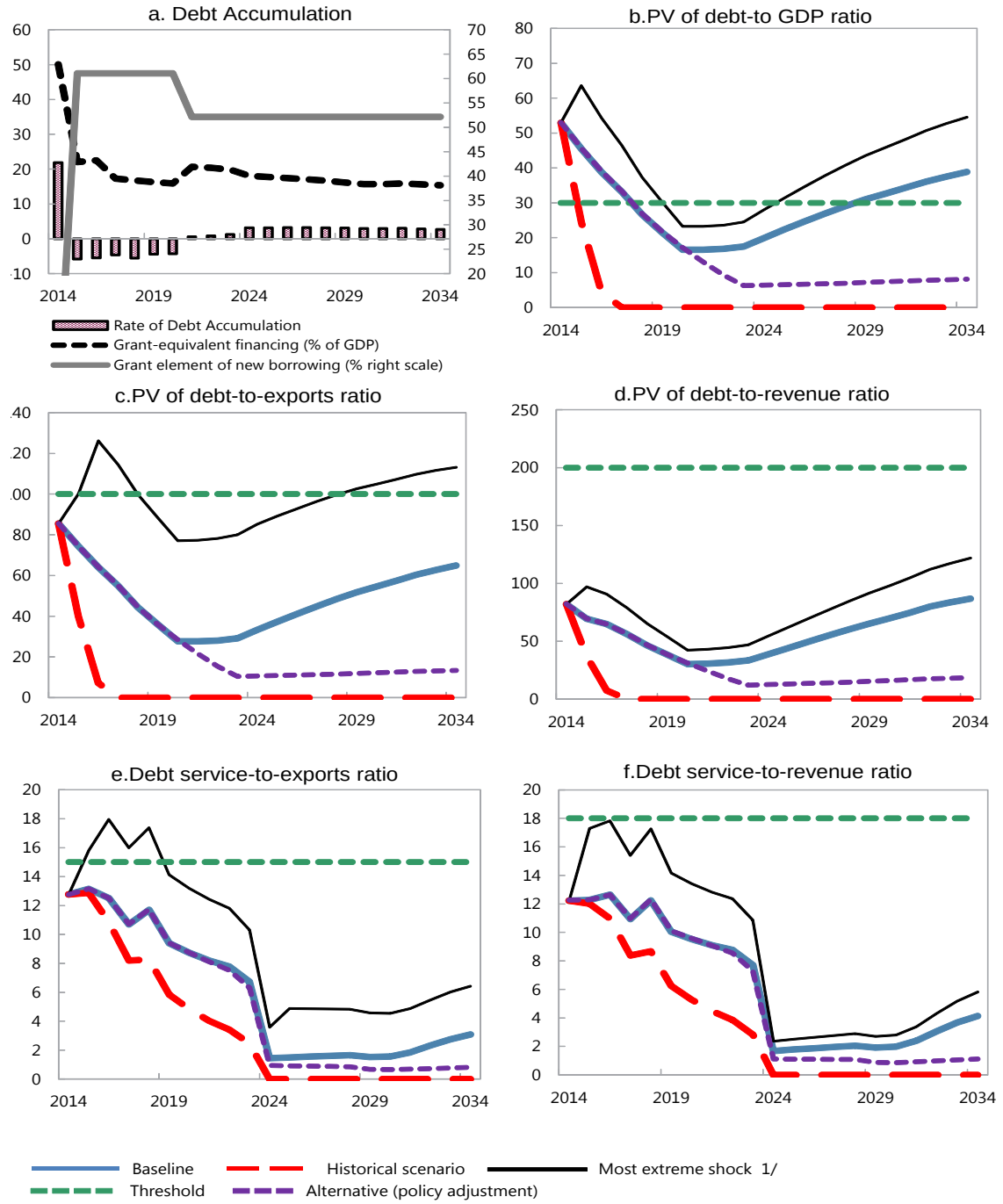
- Current account deficit will peak over the medium term at 37 percent of GDP mainly because of a spike in imports, and gradually narrow to about 4 percent of GDP over the long run. Reflecting improved capacity to tap into the marine resources by both public and private enterprises, fishing exports are expected to double in value over the next 20 years. Driven by large infrastructure projects and the purchase of fishing ships, imports would rise substantially in the next few years, and grow moderately thereafter. The trade balance will move into a deficit over the medium term, and return to a moderate surplus in 2018 and onwards.
- FDI inflows would be minimal and mostly include reinvestment of fishing joint venture earnings. Despite efforts of the authorities to privatize certain public enterprises, there has been little success in attracting foreign investors, and therefore no significant FDI inflows associated with privatization are assumed.

Fiscal Policy and Financing

- Fishing license fees are projected conservatively at around A$14 million in 2014-2019 and to gradually decline to A$11 million over the long run reflecting the adverse impact of overfishing and climate change.
- Foreign grants to the budget are expected to decline gradually from 51 percent of GDP in 2014 to 13 percent of GDP by 2034. It is assumed that the government's implementation of reforms induces continued support from development partners.
- Given the weakening fishing license fees and foreign grants, expenditures would be compressed gradually from nearly 100 percent of GDP in 2014 to 62 percent of GDP by 2034. This decline also reflects a reduced share of the public sector in the economy, which makes space for the private sector to grow.
- The overall fiscal deficit will hover around 5-7 percent of GDP over the medium term, and decline gradually to just above 4 percent of GDP by 2034. The convergence of fiscal deficit toward current account balance also points to the importance of maintaining a sound fiscal policy to safeguard competitiveness and economic stability. Financing of fiscal deficits would be mostly from external borrowing, while domestic banks suffer from severe asset quality problems and are not capable of providing sustainable financing over the long term.

1/ These expenditures would be extrabudgetary.

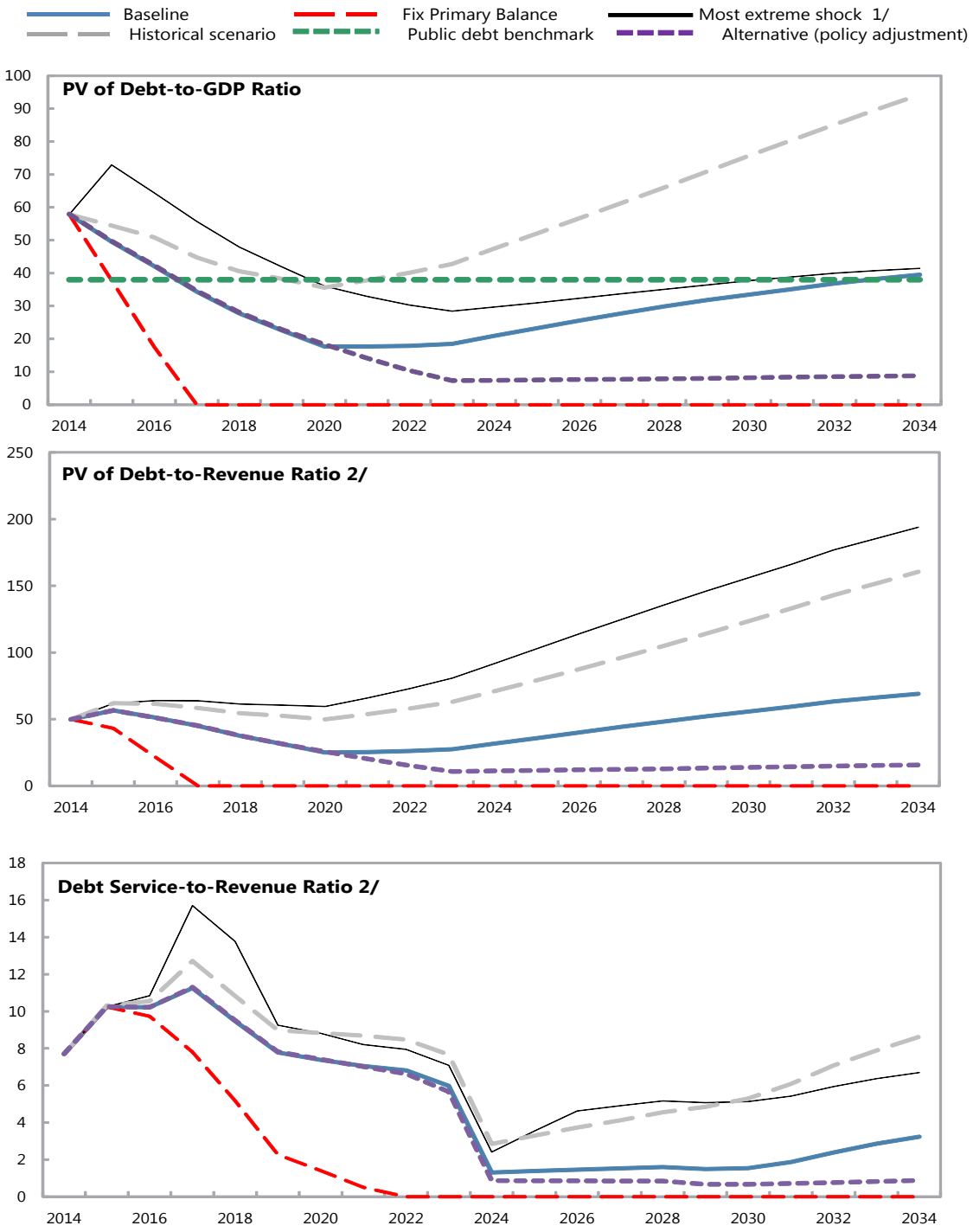
Figure 1. Tuvalu: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Alternative (policy adjustment) shock; in c. to a Alternative (policy adjustment) shock; in d. to a Alternative (policy adjustment) shock; in e. to a Alternative (policy adjustment) shock and in figure f. to a Alternative (policy adjustment) shock

Figure 2. Tuvalu: Indicators of Public Debt Under Alternative Scenarios, 2014–2034 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1. Tuvalu: External Debt Sustainability Framework, Baseline Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	40.7	36.4	35.3			52.0	44.6	37.9	32.1	25.5	20.3		35.6	68.6	
of which: public and publicly guaranteed (PPG)	40.7	36.4	35.3			52.0	44.6	37.9	32.1	25.5	20.3		35.6	68.6	
Change in external debt	-9.9	-4.3	-1.1			16.7	-7.4	-6.7	-5.8	-6.6	-5.2		5.0	2.3	
Identified net debt-creating flows	27.2	-30.1	-26.9			-30.1	34.2	20.1	16.2	6.8	8.6		3.4	1.3	
Non-interest current account deficit	34.7	-26.9	-27.9	-0.7	23.7	-29.7	34.7	20.8	16.9	7.7	9.6		5.3	3.4	4.7
Deficit in balance of goods and services	98.7	51.7	63.4			73.4	106.7	89.6	78.0	60.8	61.1		56.4	54.6	
Exports	37.8	62.8	62.9			61.9	61.1	60.5	60.2	60.0	59.9		59.9	59.9	
Imports	136.5	114.4	126.3			135.3	167.8	150.1	138.2	120.8	121.0		116.2	114.5	
Net current transfers (negative = inflow)	-37.6	-52.6	-45.4	-45.9	8.4	-55.5	-35.8	-35.3	-29.3	-22.9	-22.6		-22.9	-22.5	-22.8
of which: official	-40.9	-56.1	-48.6			-58.6	-38.9	-38.4	-32.4	-26.0	-25.7		-26.1	-25.6	
Other current account flows (negative = net inflow)	-26.4	-26.0	-45.9			-47.7	-36.2	-33.5	-31.8	-30.2	-28.9		-28.2	-28.6	
Net FDI (negative = inflow)	0.3	-4.2	-1.8	-1.6	2.3	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8		-1.8	-1.8	-1.8
Endogenous debt dynamics 2/	-7.8	1.0	2.9			1.5	1.3	1.1	1.1	0.9	0.8		0.0	-0.3	
Contribution from nominal interest rate	1.7	1.6	1.5			2.3	2.5	2.2	1.8	1.5	1.2		0.5	1.0	
Contribution from real GDP growth	-3.5	-0.1	-0.5			-0.8	-1.2	-1.1	-0.7	-0.6	-0.5		-0.5	-1.3	
Contribution from price and exchange rate changes	-6.1	-0.5	1.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	-37.0	25.7	25.8			46.8	-41.6	-26.8	-22.0	-13.4	-13.8		1.6	1.0	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	34.5			52.9	45.5	38.8	33.1	26.6	21.4		19.9	38.9	
In percent of exports	...	...	54.9			85.6	74.4	64.1	55.0	44.3	35.8		33.3	64.9	
PV of PPG external debt	...	...	34.5			52.9	45.5	38.8	33.1	26.6	21.4		19.9	38.9	
In percent of exports	...	...	54.9			85.6	74.4	64.1	55.0	44.3	35.8		33.3	64.9	
In percent of government revenues	...	...	41.6			82.0	69.4	64.8	56.1	46.4	38.3		38.8	86.8	
Debt service-to-exports ratio (in percent)	19.4	11.2	11.6			12.8	13.2	12.5	10.7	11.7	9.4		1.4	3.1	
PPG debt service-to-exports ratio (in percent)	19.4	11.2	11.6			12.8	13.2	12.5	10.7	11.7	9.4		1.4	3.1	
PPG debt service-to-revenue ratio (in percent)	15.4	12.5	8.8			12.2	12.3	12.6	10.9	12.2	10.1		1.7	4.1	
Total gross financing need (Millions of U.S. dollars)	16.6	-9.6	-8.6			-9.1	16.7	11.1	9.3	5.7	6.2		2.3	2.7	
Non-interest current account deficit that stabilizes debt ratio	44.5	-22.6	-26.9			-46.5	42.1	27.4	22.7	14.3	14.8		0.2	1.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.5	0.2	1.3	1.4	4.9	2.2	2.5	2.5	1.9	2.0	1.9	2.2	1.8	2.1	1.9
GDP deflator in US dollar terms (change in percent)	13.6	1.3	-4.9	6.6	9.1	-1.9	3.3	0.3	0.6	1.4	1.1	0.8	1.5	1.6	1.6
Effective interest rate (percent) 5/	4.1	4.0	4.0	2.7	1.4	6.4	5.1	5.1	4.8	4.8	4.9	5.2	1.7	1.6	2.1
Growth of exports of G&S (US dollar terms, in percent)	5.4	68.5	-3.5	33.8	46.9	-1.2	4.5	1.7	2.0	3.0	2.8	2.1	3.4	3.6	3.5
Growth of imports of G&S (US dollar terms, in percent)	27.7	-15.0	6.4	10.7	20.3	7.5	31.2	-8.1	-5.6	-9.7	3.2	3.1	2.6	3.6	3.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	-5.1	61.1	61.1	61.1	61.1	61.1	50.1	52.2	52.2	52.8
Government revenues (excluding grants, in percent of GDP)	47.8	56.6	82.9			64.6	65.6	59.9	59.0	57.3	56.0		51.4	44.7	49.2
Aid flows (in Millions of US dollars) 7/	8.3	11.1	9.4			19.8	9.0	9.4	7.4	7.5	7.5		11.4	13.9	
of which: Grants	8.3	11.1	9.4			19.8	9.0	9.4	7.4	7.5	7.5		7.9	9.6	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		3.4	4.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			50.1	22.1	22.5	17.3	16.8	16.3		18.1	15.4	17.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			64.8	100.0	100.0	100.0	100.0	100.0		85.6	85.4	86.1
Memorandum items:															
Nominal GDP (Millions of US dollars)	39.3	39.9	38.4			38.6	40.8	42.0	43.0	44.5	45.8		53.9	76.9	
Nominal dollar GDP growth	23.2	1.5	-3.6			0.3	5.8	2.8	2.5	3.4	3.0	3.0	3.4	3.6	3.5
PV of PPG external debt (in Millions of US dollars)			12.3			20.7	18.4	16.2	14.2	11.8	9.8		10.7	29.8	
(PVt-PVt-1)/GDPt-1 (in percent)						21.8	-5.8	-5.5	-4.7	-5.6	-4.5	-0.7	3.1	2.7	2.1
Gross workers' remittances (Millions of US dollars)	4.6	3.8	3.7			3.7	3.8	3.9	4.0	4.1	4.2		4.9	7.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	31.5			48.4	41.6	35.5	30.3	24.3	19.6		18.3	35.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	47.6			74.2	64.5	55.6	47.7	38.4	31.1		28.9	56.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	10.1			11.1	11.4	10.9	9.3	10.2	8.2		1.2	2.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Tuvalu: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	45.3	43.0	41.1			56.9	48.6	41.2	33.4	26.7	21.5		36.5	69.3	
of which: foreign-currency denominated	40.7	36.4	35.3			52.0	44.6	37.9	32.1	25.5	20.3		35.6	68.6	
Change in public sector debt	-10.3	-2.3	-1.9			15.8	-8.3	-7.4	-7.8	-6.6	-5.2		5.0	2.3	
Identified debt-creating flows	4.0	-10.7	-21.8			-19.2	2.4	4.1	6.5	5.0	4.4		4.9	2.0	
Primary deficit	7.0	-11.2	-28.3	8.9	18.7	-18.3	1.6	3.1	5.8	4.6	3.9	0.1	5.4	3.4	4.5
Revenue and grants	69.0	84.3	107.5			116.0	87.6	82.4	76.3	74.0	72.2		66.1	57.3	
of which: grants	21.2	27.8	24.6			51.4	22.1	22.5	17.3	16.8	16.3		14.7	12.5	
Primary (noninterest) expenditure	76.0	73.1	79.1			97.8	89.2	85.4	82.1	78.7	76.2		71.5	60.6	
Automatic debt dynamics	-3.1	0.5	6.5			-1.0	0.8	1.0	0.8	0.4	0.4		-0.5	-1.4	
Contribution from interest rate/growth differential	-3.4	1.1	0.7			1.0	0.5	0.4	0.4	0.2	0.2		-0.6	-1.6	
of which: contribution from average real interest rate	0.9	1.1	1.2			1.9	1.9	1.6	1.2	0.9	0.7		-0.1	-0.2	
of which: contribution from real GDP growth	-4.3	-0.1	-0.5			-0.9	-1.4	-1.2	-0.8	-0.6	-0.5		-0.6	-1.3	
Contribution from real exchange rate depreciation	0.4	-0.6	5.8			-1.9	0.3	0.7	0.4	0.2	0.2		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-14.3	8.4	19.9			35.0	-10.6	-11.5	-14.4	-11.7	-9.6		0.1	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	40.3			57.9	49.6	42.1	34.3	27.8	22.6		20.9	39.6	
of which: foreign-currency denominated	...	...	34.5			52.9	45.5	38.8	33.1	26.6	21.4		19.9	38.9	
of which: external	...	...	34.5			52.9	45.5	38.8	33.1	26.6	21.4		19.9	38.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	14.3	-3.6	-19.9			-9.3	10.6	11.5	14.3	11.6	9.6		6.3	5.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	37.5			49.9	56.6	51.1	45.0	37.5	31.3		31.7	69.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	48.6			89.6	75.6	70.3	58.2	48.5	40.4		40.8	88.4	
of which: external 3/	...	...	41.6			82.0	69.4	64.8	56.1	46.4	38.3		38.8	86.8	
Debt service-to-revenue and grants ratio (in percent) 4/	10.6	9.0	7.8			7.7	10.2	10.2	11.3	9.5	7.8		1.3	3.2	
Debt service-to-revenue ratio (in percent) 4/	15.4	13.5	10.2			13.8	13.7	14.1	14.6	12.2	10.1		1.7	4.1	
Primary deficit that stabilizes the debt-to-GDP ratio	17.3	-8.9	-26.4			-34.1	9.9	10.5	13.6	11.3	9.2		0.4	1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.5	0.2	1.3	1.4	4.9	2.2	2.5	2.5	1.9	2.0	1.9	2.2	1.8	2.1	1.9
Average nominal interest rate on forex debt (in percent)	4.1	4.0	4.0	2.7	1.4	6.4	5.1	5.1	4.8	4.8	4.9	5.2	1.7	1.6	2.1
Average real interest rate on domestic debt (in percent)	...	5.0	5.5	5.3	0.3	4.5	4.5	4.6	4.6	...	...	4.5	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciation)	0.7	-1.3	15.8	-0.9	14.2	-5.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	1.3	0.9	1.8	2.1	1.8	2.7	2.6	2.3	2.2	2.0	1.8	2.2	1.5	1.6	1.6
Growth of real primary spending (deflated by GDP deflator, in percent)	-11.7	-3.6	9.6	-21.2	65.3	26.3	-6.5	-1.9	-2.1	-2.2	-1.4	2.0	0.0	1.0	0.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	-5.1	61.1	61.1	61.1	61.1	61.1	50.1	52.2	52.2	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3. Tuvalu: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of debt-to GDP ratio								
Baseline	53	45	39	33	27	21	20	39
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	53	24	4	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2014-2034 2	53	45	39	37	33	31	44	82
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	53	48	43	37	30	24	22	44
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	53	51	54	49	43	38	36	48
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	53	48	42	36	29	23	22	42
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	53	46	39	34	27	22	20	39
B5. Combination of B1-B4 using one-half standard deviation shocks	53	44	31	24	17	12	10	35
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	53	64	54	46	37	30	28	55
PV of debt-to-exports ratio								
Baseline	86	74	64	55	44	36	33	65
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	86	40	7	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2014-2034 2	86	74	65	61	56	52	73	137
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	86	74	64	55	44	36	33	65
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	86	100	126	114	100	88	85	113
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	86	74	64	55	44	36	33	65
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	86	74	65	56	45	37	34	65
B5. Combination of B1-B4 using one-half standard deviation shocks	86	65	41	33	24	16	14	48
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	86	74	64	55	44	36	33	65
PV of debt-to-revenue ratio								
Baseline	82	69	65	56	46	38	39	87
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	82	37	7	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2014-2034 2	82	69	66	63	58	56	85	183
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	82	73	72	63	52	43	44	97
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	82	78	91	83	74	67	71	108
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	82	73	70	61	50	42	42	94
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	82	69	65	57	47	39	40	87
B5. Combination of B1-B4 using one-half standard deviation shocks	82	67	51	42	31	21	20	78
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	82	97	91	79	65	54	55	122

Table 3. Tuvalu: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	13	13	13	11	12	9	1	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	13	13	11	8	8	6	0	0
A2. New public sector loans on less favorable terms in 2014-2034 2	13	13	10	3	4	2	1	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	13	13	13	11	12	9	1	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	13	16	18	16	17	14	4	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	13	13	13	11	12	9	1	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	13	13	13	11	12	9	1	3
B5. Combination of B1-B4 using one-half standard deviation shocks	13	12	11	9	10	8	1	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	13	13	13	11	12	9	1	3
Debt service-to-revenue ratio								
Baseline	12	12	13	11	12	10	2	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	12	12	11	8	9	6	0	0
A2. New public sector loans on less favorable terms in 2014-2034 2	12	12	10	3	4	2	1	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	12	13	14	12	14	11	2	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	12	12	13	12	13	11	3	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	12	13	14	12	13	11	2	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	12	12	13	11	12	10	2	4
B5. Combination of B1-B4 using one-half standard deviation shocks	12	13	13	11	13	10	1	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	12	17	18	15	17	14	2	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	43	43	43	43	43	43	43	43

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Tuvalu: Sensitivity Analysis for Key Indicators of Public Debt 2014–2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	58	50	42	34	28	23	21	40
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	58	54	51	45	41	38	47	94
A2. Primary balance is unchanged from 2014	58	38	18	0	0	0	0	0
A3. Permanently lower GDP growth 1/	58	50	44	38	33	29	43	118
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	58	55	54	50	47	45	62	114
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	58	65	72	64	57	52	51	65
B3. Combination of B1-B2 using one half standard deviation shocks	58	61	64	58	53	50	59	95
B4. One-time 30 percent real depreciation in 2015	58	73	64	56	48	42	30	41
B5. 10 percent of GDP increase in other debt-creating flows in 2015	58	56	48	40	33	28	27	45
PV of Debt-to-Revenue Ratio 2/								
Baseline	50	57	51	45	38	31	32	69
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	50	62	61	58	54	53	71	161
A2. Primary balance is unchanged from 2014	50	43	21	0	0	0	0	0
A3. Permanently lower GDP growth 1/	50	57	53	49	44	40	63	196
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	50	62	64	64	61	60	92	194
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	50	74	88	84	76	71	77	114
B3. Combination of B1-B2 using one half standard deviation shocks	50	69	77	75	70	68	89	163
B4. One-time 30 percent real depreciation in 2015	50	83	78	73	65	58	45	72
B5. 10 percent of GDP increase in other debt-creating flows in 2015	50	63	58	52	45	39	41	78
Debt Service-to-Revenue Ratio 2/								
Baseline	8	10	10	11	9	8	1	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	10	11	13	11	9	3	9
A2. Primary balance is unchanged from 2014	8	10	10	8	5	2	0	0
A3. Permanently lower GDP growth 1/	8	10	10	12	10	9	3	10
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	8	11	11	13	12	10	4	11
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	8	10	11	16	14	9	2	7
B3. Combination of B1-B2 using one half standard deviation shocks	8	11	11	15	13	10	3	9
B4. One-time 30 percent real depreciation in 2015	8	12	14	15	14	12	3	6
B5. 10 percent of GDP increase in other debt-creating flows in 2015	8	10	10	13	10	8	2	4

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.